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A System Architecture for AI-Driven Market Entry Strategy Generation Using Large Language Models

Diana Scherz[a,b,*], Juan Antonio Ortega[a], Marcel Wieland[b], Wilhelm Daniel Scherz[a,c]

[a] *School of Computer Engineering, University of Seville, Avenida de la Reina Mercedes s/n, 41012 Sevilla, Spain*
[b] *Printec-DS HMI Solutions GmbH, Zeppelinstraße 11, 78256 Steißlingen, Germany*
[c] *Department of Computer Science, Konstanz University of Applied Sciences, Alfred-Wachtel-Straße 8, 78462 Konstanz, Germany*

Abstract

Businesses looking to expand into international markets need high-quality market research to make strategic decisions and develop effective market entry strategies. However, conducting extensive secondary market research that contains all the necessary data is very time-consuming and resource-intensive. In recent years, Artificial Intelligence (AI) has been widely used for marketing applications. Conducting market research and generating market entry strategies are functions that AI, especially Large Language Models (LLMs), could support businesses with. In this paper we propose a theoretical model of how an LLM could be trained to ensure the quality of the output by conducting secondary market research and, based on it, to generate realistic market entry strategy for decision making about international expansion. One of the key elements of the proposed pipeline is an adjustable scoring system which ensures input data reliability and transparency and, as a result, helps to provide a reliable output which could be used for strategic decision-making. Compared to traditional market research approaches, the proposed methodology offers significant improvements in speed, resource efficiency, and accessibility and is aimed to support the businesses in their expansion into new international markets.

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Keywords:
Large Language Models (LLMs); Marketing strategy; Secondary market research; Input data reliability; Data source scoring; Data quality criteria; Market entry strategy; Internationalization; International marketing; Market research; Trustworthiness in AI; Modular architecture for LLMs; Artificial Intelligence (AI); Transparency

1. Introduction

In today's rapidly evolving global environment, the necessity for swift decision-making is crucial, as asserted by Bolívar et al. [1]. This is further emphasized by Tao and Jiang [2]. Faced with the need for continued growth, com-

* Corresponding author. Tel.: +49 7738 80241-16.
E-mail address: diana.scherz@printecds.com

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panies often seek expansion opportunities in international markets. However, choosing the optimal market requires thorough research to assess a company's prospects in the available markets. Such market analysis is time-consuming because it requires collecting reliable data on many market characteristics and trends. Correctly assessing this data can play a decisive role in a company's future success in the chosen market. Additionally, the specialist conducting the research must have an understanding of the industry and possess the necessary analytical and research skills to ensure high-quality market research. They must also have strategic marketing knowledge to create a realistic market entry strategy based on the research. Considering that a company usually needs to analyse several markets before deciding to expand into one, the collected data could become outdated due to political or economic changes before the market research is complete.

Traditionally there are two types of methods which are used for market research:

- Primary market research refers to the collection of the primary data to address the problem in question [3], including quantitative methods like surveys, telephone interviews, online surveys and qualitative methods like focus groups, in-depth interviews.
- Secondary market research is based on existing datasets that have already been collected by other researchers for a different primary purpose. Such data can include official statistics, market research, analytical reports, newspaper articles, etc. [4]. Secondary market research plays an important role in market research, as it saves time and resources by using already collected data. Analysis of secondary data is used for pre-selection and rough selection of international markets, as well as premise control and strategic monitoring of potentially interesting markets [4].

The dynamic nature of today's markets and the rapid regulatory changes in various markets highlight the importance of an interactive and rapidly evolving system that helps companies expand successfully and on time. As small and medium enterprises (SME) often lack the personnel and financial resources to conduct proper market analysis, Large Language Models (LLMs) are increasingly becoming invaluable tools [5] as they streamline numerous mechanical and repetitive tasks while also aiding in the analysis of relationships to a certain degree. Artificial Intelligence are already widely used in various fields of marketing like integrated digital marketing, content marketing, experiential marketing, marketing operations, and market research [6]. Specifically LLMs are often applied to write and edit texts, edit pictures and complete simple research tasks [7].

In recent years Artificial Intelligence (AI) and Generative Artificial Intelligence (GAI) made big advances for marketing applications, which include boosting the productivity to generate marketing content [8], customer segmentation [9], and personalization of e-commerce and generation customer of insights [10]. Application of AI in marketing has shown constant growth worldwide, and Statista [11] forecasts continued growth in the future.

Artificial intelligence could also be a useful tool that allows to save resources of the company as well as increase the precision of analysis of the marketing data. However, AI is still not so widespread in marketing research and is mostly used for customer segmentation. The complexity of the tasks that modern LLMs can complete grows from year to year, and currently they show good results in collecting, summarizing and analysing the data. An accessible LLM can be configured and optimized using "verifiable data", allowing the analysis and categorization of large data sets based on predefined criteria. The foundation of both machine learning and artificial intelligence is deeply rooted in statistical analysis and the delineation of data domains.

Such development of LLMs gives rise to the idea of using LLMs to conduct secondary market research. The growing potential of LLMs to collect and summarize data for specific purposes allows businesses to reduce the time necessary for complex market research. This provides opportunities for faster, tailored market selection, reducing the time necessary for strategic decision-making in rapidly changing economic situations.

Moreover, modern LLMs could support marketers in their research activities and offer companies efficient market entry strategies based on comprehensive market research, analysis of market trends and economic data, and company priorities.

Therefore, we propose a theoretical model of how an LLM could be trained to conduct high-quality secondary market research and, based on it, offer realistic market entry strategy for decision-making about international expansion.

Such application of LLMs requires solutions to several problems to ensure the output is reliable enough for strategic decision-making:

1. High-quality, reliable, and up-to-date market research is essential for a successful market entry strategy. The quality of the data used for market research and how it is estimated is one of the key challenges for market researchers. This is why every source must be carefully verified. However, modern LLMs offer limited opportunities for precise filtering of data sources, which could lead to misleading or false results.
2. Although modern LLMs show great progress in collecting, summarizing and reasoning the data, major LLMs remain a "black box" [12]. Despite active research in the field of explainable AI, there is still no effective way to reach explainability without sacrificing the productivity of the AI. That brings us to the point where despite impressive results that LLMs can deliver, the output does not always offer the required quality. It means that the results provided by LLMs cannot be trusted completely, as it is hard to evaluate, why did LLM come to such conclusions, what data did they use to base their conclusion on or if the data was relevant, trustworthy or generally up to date. This issue becomes critical, if there is not enough market knowledge by the person who is supposed to evaluate the quality of the output.

To overcome these issues we propose an approach that would allow to moderate the input data to ensure the quality of data output.

In secondary market research, researchers typically have limited opportunities to evaluate data quality, so data source reliability becomes a primary criterion for including information in research. Using related logic, we propose developing a scoring system that labels each data source used for market research according to its estimated reliability. Based on this system, an LLM can select more reliable sources and provide higher-quality output. Additionally, integrating a data source reliability scoring system into the output will enable human specialists to more accurately estimate the "trustworthiness" of the output for further decision-making purposes. While the scoring system will be based on international academic and industrial standards and guidelines for market research and secondary data analysis, the market research and market entry strategy outputs provided by the LLM will be evaluated using real-world business data to adjust the system. The aim of this approach is to develop an application-specific, LLM-based tool that automates the process of collecting market data for reliable secondary market research and the creation of realistic market entry strategy.

2. Methodology

2.1. Process pipeline

Since this research focuses on applying and adapting existing LLMs to conduct reliable, complex research for a specific use case, we concentrate on a system that enables more control over the quality of input and output data. This ensures the quality of AI-conducted research without reducing LLM performance due to technological changes aimed at improving explainability. Thus, we propose an approach that would ensure the reliability of the data sources used by LLMs, thereby ensuring more reliable data outputs. The proposed system has the following characteristics:

- Data transparency,
- Explainability of the AI (based on controlled input),
- Comprehensible output and decision-making process of the AI,
- Tunable weights for specific use cases,
- Reliable output.

Figure 1 shows a possible methodological approach for generating secondary market research and a subsequent market entry strategy workflow. It is worth mentioning that the LLM system described here does not have to be identical and could be two different instances.

The following are the procedures and modules, along with their respective requirements:

- Step 1. Input module: this module is required to manage user input and collect the necessary information for market research and market entry strategy. The interaction is planned as follows: The user writes a request for the Market Research LLM to provide a market entry strategy for a specific company in a specific market.

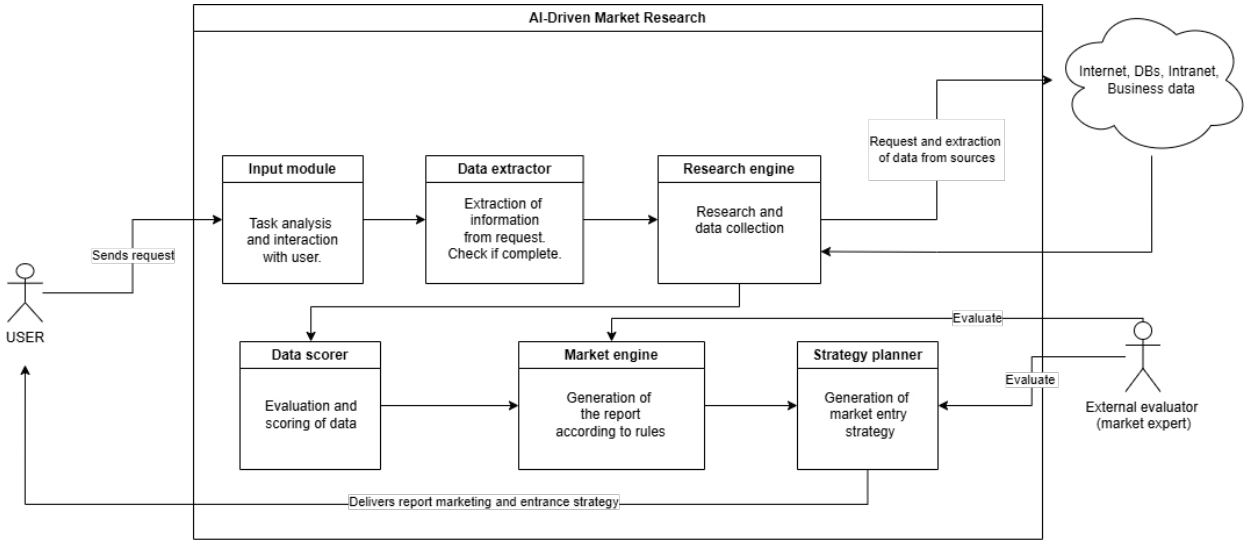

Fig. 1. Methodological approach for generating secondary market research

Important elements include product description, target market, time planning, etc. The Market Research LLM will be based on one of the existing major LLMs, as these possess outstanding analytical capabilities, providing the performance required to create an adequate market entry strategy.

- Step 2. Data extractor: this module is required for extracting the information input by the user. The module also validates that all required information has been entered. If not, it returns a request for the missing information. The LLM then processes the request. During the first iteration, the system extracts relevant information about the context, aims, and priorities of the required market entry strategy from the request and asks for additional data. There is no user interaction in this module except if there is missing information.
- Step 3. Research engine: this module is responsible for researching and collecting all the necessary data. It is important to mention that to this point the data still is not scored and that will be done in a different module. The material obtained will include datasets from various sources, such as the internet, intranet (if business data is available and accessible) and specific databases. After processing the additional information, the LLM conducts market research and extracts relevant information from various sources, such as databases, analytical reports, company websites, newspaper articles, etc. The user has no interaction in this module.
- Step 4. Data scorer: this module is one of the most important components of the system. It is in charge of processing all the data and evaluating it with the aim to score the data. This score is vital for marketing research and market entry strategy because of it depends how trustworthy dataset is. The index includes not only the dataset, but also the sources of the collected data. The scoring algorithm includes the general data reliability criteria for secondary research as well as specific criteria for market research. For example, special criteria include parameters such as how close the source is to the business, the purpose of publishing the data, and how current the data is. The higher the data's score, the more reliable the source and dataset are. Each data source is scored individually, taking into account the aim of the current research. The user has no interaction with this module.
- Step 5. Market engine: this module is responsible for generating the marketing report according to a previously defined set of rules. The use of an LLM is intended for this module. Some of the rules intended for generating the marketing report include the previously defined score from the previous module. The report fulfills marketing research report criteria according to the ESOMAR standard [13] and the data quality criteria for secondary market research formulated by Berndt et al. [14]. The LLM creates market research based on data with the highest scores. For each part of the market research, the LLM must provide an average reliability score so the user can evaluate the market research's overall reliability without a deep understanding of the market. Users do

not interact directly with this module except through the generated report, which they can read to understand the market.

- Step 6. Entrance strategy planner: this module is similar to the Market engine module will generate a report for the user but with the difference in that it generates a report for the user. However, the generated report aims to fulfill the requirements for a successful marketing strategy. This module will also use a customized LLM for generating market entrance strategies. Real strategies and the experience of specialists will be used for this. Additionally, it will be based on the market research done by the previous module and the available data. It is important to note that the previously defined score will also be used to ensure the transparency and reliability of the output. Users will not interact directly with this module except for the generated market entry strategy. During the module's training, a specialist's feedback on the generated market entry strategy will be considered.
- Step 7. User or decision maker: The user obtains the market research and the market entry strategy. Using the scores, the user can evaluate the general reliability of the result and take decision about using it in his or her business strategy.

2.2. *Input reliability control in market research*

One of the main characteristics of the system is the reliability and trustworthiness of the market research and the market entry strategy.

Market research is essential for making strategic business decisions. Fleisher and Bensoussan [15] emphasize the importance of data analysis in defining appropriate strategies: "it is only through the careful collection, examination, and evaluation of the facts that appropriate strategic alternatives can be weighted in the light of organizational resources and requirements." According to Berndt et al. [14], some of the key functions of the international market research are as follows:

- Early identification of opportunities and risks in international markets;
- Providing information to support international market decisions at strategic, tactical, and operational levels;
- Increased efficiency of international marketing planning.

Fleisher and Bensoussan [15], talking about the importance of good data inputs to analysis, mention as key criteria accuracy and reliability of sources of input data. Noting that not all data are of equal quality, Fleisher and Bensoussan emphasize the need to assess data sources to identify the originator's reasons for providing data and to recognize potential biases projected onto the data set in order to avoid incomplete or inaccurate data.

Since the purpose of market research is to support strategic decision making, it is crucial that it provides a deep and comprehensive analysis of market characteristics, trends, and parameters that might affect companies operating in the market. Among the information that is usually included in international market research is as follows (we use the categorization provided by Bauer [4]:

1. Information on global country-specific characteristics (macroeconomic situation and trends; macroeconomic structure and development; political and regulatory environment; geographical and technological conditions);
2. Information about country-specific business conditions (industry size, structure, and development; demand potential, volume, structure, and development; number, size, objectives, and strategies of competitors; number, size, objectives, and strategies of suppliers; public attitude toward the activities of foreign companies in their own country, toward the country of origin, and toward the company's products and the company itself; etc.);
3. Information on country-specific and cross-border opportunities and instruments for strategic and operational market influence (identification of partners capable of and willing to cooperate; identification of companies willing or interested in taking over; identification of lucrative market segments or niches; product, distribution, contracting, and communication policy options);
4. Information about potential country-specific market reactions, i.e., the consequences and effects that can be expected from alternative strategic and operational activities in specific situations, and information about the current reactions of those foreign markets in which the company is already active;

5. Information about the internal situation of the company (resources available for internationalization; internationalization know-how; internationalization strengths and weaknesses; etc.)

2.3. Data sources

Due to the complexity of market research and the variety of data required for a comprehensive analysis, a wide range of sources are used to collect information. Although market research literature usually mentions official statistical data and empirically based studies (e.g. country market analyses, competitor analyses, media analyses) as the main data sources, as well as reports, communications and similar publications (e.g. newspaper articles, annual reports and field reports), there is a whole range of sources that could also be used for market research, especially competitive research [16]. These sources include social media posts, online expert communities, company homepages and press releases, but they are usually not named due to their uncertain reliability. When most of the academic literature on market research was written, there was a clear distinction between reliable and unreliable sources. Nowadays, the wider usage of the internet makes access to information easier, but it is more difficult to estimate the reliability of sources. Some sources, such as government and public sector statistics and reports, as well as academic, government, and major business databases, are highly reliable. Industrial and commercial publications, including company reports, press releases, trade journals, magazines, and market research reports, are generally considered reliable sources of market information. However, the quality of information from commercial market data providers can vary. News articles and social media posts can provide valuable market insights but require careful attention to the quality of the information.

It is important to note that two types of data source are required for the system. The first type of data is required for training the Research Engine, while the second type is required for the Market Engine and Entrance Strategy Planner. Data for the Research engine will include, but not be limited to, market reports, sustainability reports, financial reports, public databases, analytical reports, company websites and newspaper articles. Data for training the Entry strategy planner will mainly consist of existing market reports and market entry strategies.

Sources of data for all three modules can include government and public sector statistics and reports, as well as academic, government and major business databases, proprietary data and web scraping. Sources of trustworthy data for market entry strategy can also come from the internet.

In the technical sense, there are many possible data sources that could be used to train the LLM, but the proposed system will initially be limited to textual data. This does not mean that data sources like video material or audio reports do not contain additional information, but most LLMs have limited capability to process this kind of information. Another reason is that the validity and trustworthiness of such media is lower by its very nature.

Modern LLMs can efficiently research a large number of data sources and collect the necessary information for market research in a short amount of time. This is much more efficient than market research conducted by humans. However, LLMs cannot always categorize data sources based on reliability as well as experienced human researchers can.

This could be improved by increasing the transparency of the LLM, which means making the LLM's decision-making process more understandable for a user. Although Explainable AI is a topic that is being actively researched, there are still various challenges that at the current stage do not allow to make AI explainable and keep its performance on the same level. However, we can address the issue of input quality in market research by improving the transparency and trustworthiness of AI's decision-making process. This can be achieved by establishing rules that help the LLM filter data sources more efficiently and avoid bias. This will improve the quality of the market research output without affecting the LLM's performance and allow us to leverage the full advantages of AI for strategic decision-making. To achieve this, we propose a scoring system that categorizes data sources according to their reliability. The scoring system's main principle is to help the LLM simulate the way human researchers evaluate source reliability.

2.4. Data quality criteria

Data quality criteria for secondary market research include Actuality, Quality, Comparability, Decision relevance, and Cost-effectiveness [14]. The proposed scoring system/ranking model mostly evaluates two of these criteria: Quality, which includes objectivity, reliability, and validity of the collected data, and Actuality. In accordance with these fundamental principles, a more comprehensive system of subcriteria must be established to ensure the complex and

accurate estimation of data quality. These subcriteria have yet to be defined. In this system, each subcriterion is a variable with a specified weight demonstrating the extent to which market research data aligns with the established criteria. Based on the values of all the variables, data sources will be clustered so that the LLM considers the data sources with the highest scores for the market research, facilitating further evaluation and analysis of the output. The output will include the average reliability score for each part of the market research and the average score for the complete market research so that users can evaluate the quality of the sources used and, based on that, take the decision about the trustworthiness of the market entry strategy.

2.5. Training and test

Real business data from internal sources will be used to evaluate and fine-tune the scoring system and the prompt. This data will include detailed information about the market, including human-made market research, as well as internal business data from companies already operating in this market. Based on this data, we will evaluate the accuracy of the market research and the proposed market entry strategy for the chosen market. After evaluating the input, the scoring system and prompt will be analysed and adjusted to improve the quality of the output. Thanks to that architecture we can compare different fine-tuning techniques to find one which offers the optimal performance for the use case. Generally speaking, there are the following methods to fine-tune an LLM: zero-shot, few-shot, task-specific fine-tuning, instruction fine-tuning, multitask and sequence-to-sequence [17].

The training phase will also involve selecting the most effective prompting technique. We plan to compare the performance of zero-shot and few-shot techniques. However, we expect the few-shot technique to perform better, since in this case we only need to provide the model with a small amount of examples to help it understand the task better. As applied in similarly challenging fields, few-shot can outperform zero-technique [17].

This will be followed by a testing phase in which the prompt and scoring system will be used to create market research and a market entry strategy for a market with no real business data. Market experts will evaluate the quality of the output using the Delphi method. The Delphi technique is a qualitative research method widely used in the social sciences for identifying and analysing managerial decision-making processes. It involves gathering insights and forecasts from a panel of experts through a series of structured questionnaires, which are usually conducted in multiple rounds. After each round, the facilitator provides summaries of the experts' forecasts and rationales in an anonymous format, enabling the experts to revise their views in light of the group feedback. This iterative process continues until a consensus or stable pattern of opinions emerges. The aim is to use expert judgement to reach more accurate or reliable conclusions in situations where hard data is limited or uncertain. [18]. The training phase will also entail selecting the optimal prompting technique. This will be followed by a testing phase in which the prompt and scoring system will be used to create market research and a market entry strategy for a market with no real business data. Market experts will evaluate the quality of the output using the Delphi method. This evaluation aims to analyse the reliability of the input data used by the LLM, as well as the general reliability and trustworthiness of the market research and market entry strategy. If the output is not satisfactory, further fine-tuning or changes to the scoring system will be necessary.

Sometimes, LLMs generate plausible yet false content as output, a phenomenon known as hallucination [19, 20]. For example, in the case of market research and the generation of market entry strategies, market players may be mentioned several times with different or incorrect financial figures, retailers may be treated as manufacturers, and the countries in which multinational players operate may not be specified; any nuances that could play an important role in creating the strategy may also be overlooked. Such output, especially in the context of market research and market entry strategy can mislead strategic decision-making and cost companies significant financial losses. A catalogue of criteria is to be designed to estimate the quality of the output. These criteria must provide a comprehensive evaluation of the market research and the market entry strategy based on it. Groups of criteria will, for example, estimate the completeness of the information and estimated market parameters, the actuality and reliability of the information, the correctness and completeness of the analysis, the adequacy and executability of the market entry strategy, and its integrity. This catalogue of criteria will be used to evaluate the accuracy of the output during both the training and test phases. Market experts who evaluate the output will also be invited to base their decisions on the catalogue of criteria.

3. Expected results

The proposed model is expected to improve market research activities and the creation of market entry strategies, while also improving the quality of input data. It will provide extensive research, significant time savings, and analysis of hidden dependencies that are often invisible to human market researchers. The input reliability scoring system manages biased input data without reducing the performance of LLMs. This paper's proposed model can significantly improve strategic decision-making processes in international business expansion, helping companies save costs and reduce risks when entering international markets. The model requires combining theoretical research on parameters that ensure input data quality with real-world business data to ensure the quality of the system's output. However, the scoring system for filtering data sources according to their reliability can be used not only for LLMs, but also for human-conducted market research, as it provides a general system of criteria to help decide the reliability of a data source.

4. Discussion and further research

The constant development of AI leads to the steady growth of possible applications of AI-based tools, as well as growing business interest in these tools. AI-based tools can significantly improve the efficiency of research-intensive tasks by decreasing the time required to complete them, improving the depth of the research, and enhancing the quality of the subsequent analysis. This frees up employees to focus on strategic decision-making.

However, the current level of explainability of LLMs makes it difficult to rely on the results of AI-conducted research for strategic decisions. At this stage of technological development, it is important to develop human-controlled filtering systems to ensure the quality of input data. A proposed scoring system for input data sources could be one possible solution.

Although the exact LLM to be used has yet to be chosen, evidence suggests that modern state-of-the-art LLMs do not demonstrate significant performance differences on specific tasks [21]. However, this question still needs to be clarified for this specific use case.

Combining theoretical approaches to secondary research, practical experience in market research, and the research and analytic capacities of LLMs trained on real business data can provide a powerful tool for supporting strategic decision-making. In this sphere, high-quality research is vital for reducing financial risks. However, careful selection and evaluation of parameters for the scoring system, as well as precise fine-tuning of the model, are required to ensure delivery of the best possible results and avoidance of bias.

The scoring system could be expanded for further business applications of AI-based tools, where output quality depends on input quality. We see great potential in developing input filtering systems for LLM-based research and analysis tools to improve their transparency and explainability.

5. Conclusions

The rapid development of AI, especially LLMs, in recent years has provided businesses with various opportunities to increase the efficiency of their marketing processes. Although AI is already used in many marketing domains, there are still fields where the potential for AI adoption has not been realized. One such field is conducting secondary market research and generating market entry strategies. We propose a pipeline and a scoring system that would overcome the challenges of input data reliability by applying LLMs to this task, thereby ensuring the quality of the output. The advantages of the proposed methodology are: 1) transparency of the input data, 2) partial explainability of the AI algorithm made possible by controlled input, 3) improved comprehensibility of the output and AI decision-making strategies, 4) a scoring system that can be adjusted for specific use cases, and 5) reliable output that can be used directly in strategic planning. Further research on this topic will enable the creation of a tool that supports strategic decision-making for businesses interested in internationalization by providing extensive, high-quality market research and realistic market entry strategies.

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